AJ MENDES

SOFTWARE ENGINEERING | SECURE SYSTEMS | TESTING

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PROFILE

Computer science student building secure, testable applications across Python, Java, and TypeScript. Comfortable owning protocol design, application state, user interfaces, automated tests, and the documentation needed to explain engineering decisions.

EDUCATION

Colorado School of Mines

B.S. Computer Science | Minors in Military Science and Quantum Engineering | Cyber Defense Education Certificate | Expected May 2027 | GPA: 3.5

SELECTED PROJECTS

Secure Chat Application

Python | TCP sockets | cryptography | Tkinter

- Built a Python TCP client-server chat application using RSA-2048/OAEP key exchange and AES-256-GCM authenticated encryption; the relay forwards ciphertext without access to plaintext.
- Implemented persistent keys, a framed message protocol, Tkinter UI, tamper detection, and component test scripts for crypto, protocol, client, and server behavior.

ClueGame

Java | Swing | JUnit 5 | object-oriented design

- Developed a Java Swing implementation of Clue with configuration-driven board loading, graph-based movement, card and rule state, and computer-player decisions.
- Validated game behavior with 80 JUnit 5 tests covering adjacency, targets, dealing, suggestions, accusations, and exception paths.

Call for Fire Trainer

React | TypeScript | Vite | browser APIs

- Created a browser-only React and TypeScript training application with structured command parsing, scenario generation, guided correction loops, scoring, and saved session history.

TECHNICAL EXPERIENCE

Undergraduate Data Science Researcher

The Data Mine, Purdue University | Jan 2026 - May 2026

- Built Python evaluation tools for K-means clustering on labeled synthetic cybersecurity data, reporting accuracy, adjusted Rand index, and normalized mutual information across controlled experiments.
- Developed centroid-matching and drift-visualization workflows using PCA, UMAP, and t-SNE; documented and presented findings with Purdue, Data Mine of the Rockies, and CrowdStrike collaborators.

Electrical Control Panel Technician

Xpect Solutions, Longmont, CO | Jun 2025 - Aug 2025

- Assembled and wired industrial electrical control panels from schematics, wiring diagrams, and technical drawings.
- Routed, bundled, terminated, and labeled conductors to NEC and company standards while performing detail-focused quality checks across concurrent builds.

Construction and General Labor

Formit Right Concrete, Longmont, CO | Summers 2020 - 2024

- Completed five summers of full-time concrete, site-preparation, material-handling, and landscaping work in changing jobsite conditions and early-shift schedules.

SKILLS AND LEADERSHIP

Languages: Python, Java, C++, SQL, R, TypeScript/JavaScript, HTML/CSS

Engineering: OOP, data structures, TCP protocols, cryptography, testing, Git, Linux

Leadership: Army ROTC, Mines ORESEC/CCDC, Secret security clearance